SNEHAL LAL DAS

ML Engineer · GenAI Developer · MLOps Engineer
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PROFESSIONAL SUMMARY

Final-year CS (AI & ML) student with a track record of building and shipping production ML and agentic AI systems independently. Architected a 270M-parameter Transformer from scratch, deployed a live AI-powered security tool (CodeSentinel), built a production agentic interview coach (JobGenie AI), and engineered an end-to-end MLOps pipeline with drift detection, CI/CD, and real-time monitoring. Seeking full-time ML Engineer / GenAI Developer roles where I can own the full model-to-production lifecycle.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
ML / DL: PyTorch, Scikit-learn, Transformers, CUDA, HuggingFace (familiar)
GenAI & LLM: RAG, Prompt Engineering, Context Engineering, Embeddings, Vector Databases, ChromaDB, Mistral AI, Mistral Codestral, Semantic Chunking, Multi-Agent Systems, MCP (Model Context Protocol), LangChain (familiar), Fine-tuning (familiar)
MLOps & DevOps: Docker, Kubernetes, Jenkins, Apache Airflow, CI/CD, Git, Linux
Monitoring: Prometheus, Grafana, Alibi-Detect, Redis
Backend: FastAPI, Flask, REST APIs, JWT, PostgreSQL, SQLite, SQLAlchemy
Cloud & Deployment: Google Cloud Platform (GCP), Render, Vercel, GitHub Pages
Data: Pandas, NumPy

PROJECTS

JobGenie AI — Agentic Interview Coach Jan 2025 – Present
Stack: Python · FastAPI · PostgreSQL · ChromaDB · Mistral AI · Next.js · Vercel · Render GitHub ■

- Designed and deployed a production agentic interview coaching platform with multi-turn dialogue, resume-aware question generation, and AI-powered answer scoring using Mistral AI.
- Built a RAG pipeline over user resumes and job descriptions using ChromaDB and semantic embeddings, enabling context-grounded, role-specific interview sessions.
- Engineered a secure multi-user FastAPI backend with PostgreSQL, JWT auth, and full data isolation; resolved critical schema bug (missing user_id) that caused cross-user data leakage.
- Deployed frontend on Vercel and backend on Render with UptimeRobot-based keep-alive to eliminate cold-start latency; integrated token-cost-aware Mistral inference for scalable scoring.

CodeSentinel — AI Code Vulnerability Scanner Feb 2025 – Apr 2025
Stack: Python · FastAPI · Mistral Codestral · SQLite · JWT · GitHub Pages GitHub ■ | Live Demo ■

- Architected and shipped a production-grade AI security tool that detects 4+ vulnerability classes (SQL injection, hardcoded secrets, missing validation, insecure auth) across 8 programming languages using Mistral Codestral.
- Reduced manual code review effort by automating severity triage — classifying findings into Critical / High / Medium / Low with concrete, per-finding fix suggestions.
- Engineered a secure backend with JWT authentication, bcrypt password hashing, and a full scan-history REST API; achieved zero reported auth vulnerabilities post-deployment.
- Owned the full product lifecycle — frontend (GitHub Pages), backend (Render), and CI pipeline — delivering a publicly accessible tool solo in under 3 months.

Codebase RAG System Jan 2025 – Mar 2025
Stack: Python · ChromaDB · Mistral AI · sentence-transformers · FastAPI GitHub ■

- Designed and deployed a production-grade RAG pipeline enabling natural-language querying across large multi-file codebases, reducing codebase navigation time by an estimated ~60% vs. manual search.
- Increased retrieval precision by ~40% over naive line-based splitting by implementing language-aware semantic chunking via AST parsing (Python) and regex (JS/TS).
- Optimised vector search with ChromaDB and sentence-transformer embeddings to achieve sub-2s end-to-end query latency; integrated Mistral AI for contextual answers with file-path and line-level citations.

Custom Transformer Language Model — 270M Parameters Apr 2024 – Jun 2024

Stack: PyTorch · Python · CUDA · T4 GPU GitHub ■

- Architected and trained a 270M-parameter Transformer LLM from scratch in PyTorch implementing Gemma-style architecture: GQA, RoPE positional embeddings, RMSNorm, and gated FFN.
- Maximised GPU throughput to ~100% T4 utilisation (~9 GB VRAM) via gradient accumulation and mixed-precision (AMP) training, sustaining ~8,000 tokens/sec across 29,000 training steps.
- Achieved validation loss of 2.3083; built a resume-capable checkpointing system and deployed the trained model via a FastAPI inference API with CPU-compatible serving.

EDUCATION

BTech — Computer Science Engineering (AI & ML)

Roorkee Institute of Technology, Uttarakhand, India

CGPA: 6.68 / 10 (through Sem 6) · Relevant coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Systems, Cloud Computing

Sep 2022 – Jun 2026 (Expected)

CERTIFICATIONS

- TensorFlow for Deep Learning Bootcamp — Udemy
- MLOps on GCP: CI/CD, Kubernetes, Jenkins — Udemy
- Introducing Generative AI with AWS — Udacity